The International Treaty
ON PLANT GENETIC RESOURCES FOR FOOD AND AGRICULTURE

Full Project Proposal Format

Third Call for Proposals under the Benefit-sharing Fund

Deadline for submitting full project proposal: 5th of December 2014
at Treaty-Fund@fao.org and PGRFA-Treaty@fao.org

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PROJECT PROPOSAL COVER SHEET

Project No. _____ (*For Treaty use. Do not write anything here)*

Project Title: An Integrated Approach to Identify and Characterize Climate Resilient Wheat for the West Asia and North Africa Region

Project duration: 36 Months

Target crops: Wheat (*Triticum spp*.)

Targeted developing country/ies:, Egypt, Ethiopia, Jordan and Sudan

Other Contracting Party/ies involved:

Ethiopian Institute of Agricultural Research (EIAR)- Ethiopia, National Center for Agricultural Research and Extension (NCARE)- Jordan, Agriculture Research Center (ARC)- Egypt, Agricultural Research Corporation (ARC)- Sudan

Project geographic extension (km²): 296,540.

Total requested funding: 500,000 $

Total co-funding: 500,000 $

Please select the type of project you are applying for:

☐ Single-country Immediate Action Project (Window 2)
☐ Multi-country Immediate Action Programme (Window 2)
☐ Single-country Co-development and Transfer of Technology project (Window 3)
☒ Multi-country Co-development and Transfer of Technology project (Window 3)

Applicant

Name of Organization: International Center for Agricultural Research in the Dry Areas (ICARDA)

Type of organization: CGIAR Center

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GENERAL REQUIREMENTS

These guidelines have been prepared to support applicants in the development of full project proposals. They describe the requirements that all applicants should adhere to when developing their full project proposal.

Please make sure you read these guidelines carefully before proceeding to fill in the Project Proposal Form. The full proposal should be prepared taking into account the thematic focus of the Third Call for Proposals, including in particular, the rationale, scope and expected outputs for each Window and sub-Window.

Project proposals must be clear and realistic on the problem to be addressed and objectives it tries to achieve. Project objectives have to fit in the thematic focus of the call and ultimately contribute to food security and poverty alleviation. Project objectives have to be logically interlinked with the planned activities, outputs and expected outcomes. The objectives and outputs have to be feasible in terms of duration and resources requested. The information to be provided in each section has to be focused and straightforward, qualitatively and quantitatively measurable in terms of what will be done, with what purpose, who, why and how will be involved in the activities to be implemented, who and how many will directly and indirectly benefit from the implementation of the project. A good full proposal will have a sound, clear and logically linked methodology of implementation and management.

The full project proposal should contain no more than fifteen (15) pages of text (Appendixes, table of contents and cover sheets excluded). The number of pages allocated to each section is a guide. The information required can be less but not more than the pages stipulated. All Appendixes should be duly filled in according to the provided guidelines as they form an integral part of the full project proposal. Project proposals lacking at least one Appendix, will be excluded from the selection process. The Appendixes will be provided to you in separate files together with the present document.

When submitting the full project proposal, additional attachments (endorsement letters, funding commitments, certification of the status of the organization) can be provided.

Please ensure that the project proposal and all attachments are legible in Times New Roman 12 and provided in two formats (pdf and word). Make sure the signature of the project coordinator is put on the signature page.

The project proposal, if approved for funding by the Bureau of the Sixth Session of the Governing Body, will form an integral part of the contractual agreement (Letter of Agreement) that will be signed with each applicant organization of the approved projects.

SECTION A: EXECUTIVE SUMMARY

1. Executive summary

Climate change is expected to cause adverse effects to agriculture around the globe, with many countries showing increasing concerns about their possible devastating ripercussions. Earth temperatures are expected to increase by approximately 2 ˚C in the next 100 years, which will result in dramtic changes in rainfall patterns in many parts of the world and particularly in dry areas. Temperature and water availability are major determinants of plant productivity. Limited water supply (drought stress) and warmer temperatures (heat stress) will certainly reduce crop production. Wheat yields are expected to drop by approximately 5% at each 0.5 ˚C increase in temperature.

Wheat (*Triticum aestivum* L. and *Triticum durum*) is the third most cultivated crop on the planet and the first for human uptake of dietary proteins and fibers (FAOSTAT 2012). Modern wheat cultivars were developed from a genetically narrow germplasm by means of pedigree. This increased its vulnerability to climate change, stress, and diseases. Decrease in potential wheat yield due to climate change stresses is very likely the outcome of the inability of the modern variety to adapt to hotter and/or shorter growing periods and prolonged water deficit conditions and heat stress. Such conditions are already very explicit in most of the dry areas of the World, especially within West Asia and North Africa (WANA) region. Furthermore, in most of the countries of WANA, wheat production is currently not sufficient to meet the rising domestic demands by the growing populations. Millions of tons of grain are imported annually to satisfy these needs, which raises the question of what will happen when climate change will cause a reduction in the global production and therefore to the exportations.

Wheat landraces, with high level of adaptation to drought and heat stresses, are rapidly disappearing from the main wheat production areas in the targeted countries and are replaced by new genetically uniform cultivars. Therefore, there is an urgent need to collect such wheat material and to characterize it at the molecular and phenotypic levels to enable their utilization in future breeding strategies to cope with climate change. Collected wheat genetic material deposited in ICARDA and national genebanks is not fully exploited (less than 8% are utilized) and their potential as a source of new alleles for breeding for drought and heat stress tolerance is not fully utilized. Therefore, there is a need to study the durability and variability in such material, which have been selected over millennia and that might be adapted to local harsh environments. This will lead to the identification of new alleles for tolerance against drought and heat stress, which will be utilized in ICARDA and National Agricultural Research Systems (NARS) breeding programs.

In this study, an integrated approach that includes different molecular, phenotypic and informatics tools, will be adopted to identify and characterize wheat genetic resources resilient to climate change with special emphasis on drought and heat stresses. The study will seek the establishment of an effective long-lasting partnership between ICARDA, as a leading institute, and NARS in targeted countries. This will provide the breeders and other scientists at ICARDA and NARS with the proper tools to share and integrate phenotypic and genetic information to accelerate the delivery of new wheat varieties that are adapted to the climate change-related challenges with particular emphasis on information exchange, technology transfer and capacity-building.

SECTION B: PROJECT DESCRIPTION AND CONTENTS

2.1. Problem definition

Climate change and associated conditions are growing concerns affecting many regions and earth temperature is expected to increase causing changes in rainfall patterns in many parts of the world (Thornton et al., 2014). The worst climate change scenarios envisage major impact on food production and food insecurity implying direct effects on agriculture in many countries around the globe (Challinor et al., 2014). In West Asia and North Africa (WANA) region, the dryland farming systems are expected to be severely affected by climate change (Thomas 2008) and projected to reduce renewable surafce and ground water reseources with inceased temperature in most dry regions (IPCC, 2014). Temperature and water availability are major determinants of plant productivity and, under climate change, limited water supply and warmer temperatures will most probably reduce the yield of any given crop. Such new conditions, associated with climate change, will have major negative impacts on crop productivity of many cereal crops, including wheat, adding more suffering to the growing populations and poor-resource farmers in dry areas.

The WANA region is considered among the largest food-deficit regions in the world where large scale importation of grains, particularly wheat, is common. Future scenarios predict an increased dependency on food importation, thus worsening the food security situation in the region. Besides, countries with high population growth rates will face natural resource degradation resulting in facing high rates of food insecurity. Taking Ethiopia as an example, the country is among the poorest countries in the world, with a per capita income of $470. Approximately 40 percent of the population still lives below the poverty line and life expectancy of the population at birth is only 56.19 years (World Bank 2014). To improve the food security in the country, there are governmental plans to expand wheat cultivation into new areas across the country. However, growing wheat in such areas will require cultivating high yielding and drought and heat tolerant varieties. Similar plans are being made by Egyptian government to boost wheat production in the new valley and Toshka areas in southern parts of Egypt. Such areas are prone to climate change associated conditions, particularly drought and heat stress. In Sudan, which imports 80 percent of its annual wheat needs, there are plans to boost wheat production by increasing the cultivated area. Climate change scenarios estimate a crop yield reduction in Sudan by over 20% due to heat stress (Jones and Thornton, 2003). Based on the water stress index (the value of annual rainfall divided by total population), the availability of water in Jordan is classified as “absolute scarcity” (Mudabber, 2007). The impacts of dry conditions associated with climate change on rainfed agriculture in Jordan is clear and include the complete failure of crops in the field and increased cost of providing extra feed to animals on smallholder-farmers. Drought will affect the dryland mixed systems by excluding wheat and replacing it with other crops.

Many models predict that climate change will have major negative effects on wheat production areas, especially in the WANA region (Habash et al., 2009). This reduction in yield is expected due to temperature increase, uneven distribution of rainfall and the shift toward short growing seasons. Therefore, a major challenge for wheat improvement in WANA region is how to plan for future climate change worst scenarios. Under climate change conditions, wheat plant will require complex tolerances to the extreme pressures of drought and heat (Habash et al., 2009). In a recent study, a process-based model to predict wheat adaption to the detrimental effect of climate change indicted that the negative impact on yield through climate change would be decreased and mitigated due to the adoption of new short growing and faster maturating varieties with enhanced tolerance to drought and heat stresses (Semenov et al., 2014). In another recent study, a data set based on more than

1700 simulation models indicates a negative impact of climate change on wheat yield in different regions around the globe if no adaptation measurements are considered (Challinor et al., 2014). However, the same data set predicts a yield increase, 8 to 15%, if more effective climate change adaptive varieties were developed. Therefore, tolerance to heat and drought stresses should remain a priority for the genetic improvement of wheat in dry areas.

Under climate change conditions, the adaptation of new wheat varieties is needed to cope with yield reduction associated with heat and drought stresses. To cope with these new conditions, there is a need to breed for new high yielding wheat varieties with improved resilience to climate change and particularly drought and heat stress. Efficient and effective genotypes with a short growing season and tolerance to major abiotic stresses should be utilized to produce new improved cultivars with enhanced productivity to overcome the adverse effects of climate change (Semenov and Halford, 2009). Therefore, improving and upgrading capacities of current wheat breeding programs at ICARDA and NARS in WANA region through the utilization of wheat genetic resources and modern molecular breeding tools are required to accelerate the delivery of such varieties. In addition, breeders need to use techniques from single selection to complex molecular methods to integrate desirable traits into existing varieties to improve wheat productivity in production areas vulnerable to climate change (Araus, et al., 2008). Out-scaling current selection technologies to national partners through training will help breeders select new material with desired traits for their programs.

Based on the frame conditions and challenges encountered due to climate change, the proposed study will focus on new emerging technologies and information tools to utilize genetic resources for the benefits of gene banks and wheat breeding programs in WANA region. This project will provide the breeders and other plant scientists, as direct beneficiaries, at ICARDA and NARS with the proper tools, and phenotypic and genetic information to accelerate the delivery of new wheat varieties that are adapted to the climate-related challenges in WANA regions. Furthermore, the proposed study will utilize plant genetic resources from ICARDA and national wheat breeding programs in WANA region besides the identification of putative wheat genetic resources at ICARDA, NARS and other gene banks that are resilient to climate change associated stresses (with particular emphasis on drought and heat tolerance) using the Focused Identification of Germplasm Strategy (FIGS) aligned with WANA geographical and climatic data available from the newly developed ICARDA Geoinformatics Portal (http://geoagro.icarda.org/).

This special set of wheat germplasm (elite lines, breeding material and landraces) will be assembled and managed by the ICARDA gene bank team and will be subject to genetic diversity analysis using a defined set of SNP markers (~50 markers) that will include already developed diagnostic markers for major agronomical traits (plant height, flowering time) and major pathogens resistance genes (rusts). Two selected subsets of a total of ~250 durum and ~250 bread wheat accessions will be evaluated in research stations in the targeted countries for their performance under heat and drought field conditions. To identify marker associations with heat and drought tolerance, the two subsets will be genotyped with a suitable robust and high-density SNP genotyping platform. The identified climate change resilient wheat lines and the molecular data of SNP markers associated with improved performance under heat and drought tolerance in targeted areas will be made available to the national program for future breeding applications.

In addition, all genotypic, phenotypic, collection sites and related data of the characterized wheat material will be integrated and used to develop a web-based tool to help ICARDA and national programs and scientists in WANA region to facilitate future selection in their breeding programs. The new database will be designed as an open access for data (molecular and phenotypic) sharing and deposition to the benefits of all national

institutions in the region and the world (taking into consideration the legislation and regulations regarding intellectual property protection) with options to enable new data deposition and linkage with other existing platforms.

Finally, the proposed study will include a capacity development component including training of human resources from the key national partners on the use of molecular markers in wheat breeding programs (with special emphasis on new genotyping technologies for large scale analysis) and the resulted information management systems beside already existing gene bank management systems.

2.2. Project objectives: Overall and specific objectives

The overall goal of this study is to identify and characterize climate-resilient wheat germplasm that can be used to develop high yielding varieties with improved adaptation to the drought and stress conditions prevalent in the WANA region.

The identification and utilization of PGRFA to develop new wheat lines with improved tolerance to drought and heat stress conditions prevalent in WANA region is needed for better food security and sustainable productivity. This will eventually improve the productivity of wheat in the targeted region, increase the livelihood of low income communities and resource-poor framers in WANA region and contributes to food security in targeted countries.

In addition, the proposed study aims to enhance the current capacities of NARS breeders, to promote partnerships and participation and to ensure knowledge sharing with the wheat scientists in the region.

The project specific objectives are:

1- To evaluate a large collection of genebank accessions for two years, under contrasting drought and heat field conditions.

2- To identify novel wheat genetic material with improved tolerance to drought and heat stresses.

3- To identify useful alleles for breeding drought and heat stress tolerance by means of association mapping.

4- To use molecular markers for simplifying the transfer of useful alleles associated with drought and heat tolerance into ICARDA and national breeding programs.

5- To improve the current capacities of the conventional wheat breeding programs in the region by using cutting-edge molecular and breeding tools.

6- To develop a web-based tool for integrating data at the accession level with genomic and phenotypic information with accessibility to all institutions in WANA region.

7- To provide training courses in targeted countries, workshops and scientific publications.

2.3 Targeted outputs, activities and related methodology of implementation

Output 1: *Climate change resilient wheat genetic resources with improved tolerance to drought and heat stress are identified and conserved.*

Target: Assembly of a wheat genetic resources special set that include climate resilient, drought and heat stress tolerance wheat accessions and germplasm from ICARDA and NARS breeding programs and gene banks.

Activities:

1. Identification and multiplication of wheat accessions from drought and heat stress prone areas using an improved FIGS approach.
2. Utilization of wheat breeding material and populations from ICARDA, international and national breeding programs.

Outcomes: ~1250 diverse wheat accessions are characterized, multiplied and conserved.

Output 2: *Two diverse subsets that include climate resilient, drought and heat stress tolerance durum and bread wheat genotypes identified using improved FIGS and modern genomics tools.*

Target: Genetic diversity and population structure analysis of the wheat special set using diagnostic SNP markers linked to growth habit, plant height and other related traits to produce two subsets (bread and durum wheat).

Activities:

1. Testing ~50 SNP markers on ~1250 wheat accessions from the special collection set using the KASP® genotyping assays.
2. Genetic diversity and population structure analysis to produce two subsets for bread and durum wheat genotypes for genome wide association mapping studies.

Outcomes: Two subsets for bread and durum wheat that contain at least 250 diverse wheat accessions per species are characterized using molecular markers available.

Output 3: *Field evaluation of the two subsets for yield potential and stability under drought and heat stress conditions.*

Target: Identification of wheat lines with improved tolerance against drought and heat stress and phenotypic data to execute a genome wide association study.

Activities:

1. Evaluation of the two AM panels (250 for each species) for their performance under drought and heat stress conditions.
2. Phenotypic data collection and analysis for major yield and yield components traits associated with drought and heat stress tolerance in the two panels.

Outcomes: Phenotypic data for the field performance of the two subsets under drought and heat stress tolerance available.

Identification of at least 20 wheat lines from each species with improved tolerance against drought and heat stress.

Output 4: *New alleles for drought and heat tolerance in the two subsets are identified using a genome wide association mapping approach.*

Target: Identification of molecular markers linked to traits conferring drought and heat stress tolerance in the two subsets.

Activities:

1. Genotyping of the two subsets using high-throughput SNP genotyping systems.
2. Genome wide association mapping analysis using genotypic and phenotypic data from the field evaluation experiments.

Outcomes: Identification of at least 10 QTLs against drought and heat stress in each subset.

Output 5: *Existing and newly developed SNP markers for selected agronomical traits, and abiotic stress tolerance are utilized in ICARDA and NARS wheat breeding.*

Target: Marker assisted selection using high-throughput genotyping platforms are deployed in ICARDA and NARS breeding programs.

Activities:

1. Conversion of at least 10 SNP markers associated with drought and heat stress tolerance into KASP® assays.

2. Screening at least 500 lines from ICARDA and NARS wheat (durum and bread) breeding programs using at least 20 SNP markers associated with drought and heat tolerance through the KASP® genotyping assays.

Outcomes: Implementation of a marker assisted selection system for selected traits in ICARDA and NARS breeding programs using SNP markers and high-throughput genotyping platforms.

Note: The activities of this particular output might be extended after the end of the third year (year four in Appendix 3) as a continuity of impact operations of the proposed study.

Output 6: *Open access integrated database with accession passport information, genomics and phenotypic data*

Target: Development of user friendly web-based tools to facilitate data and information exchange between participants.

Activities:

1. Database development and incorporation of field and molecular data, results and markers information for wheat material used in this study.
2. Database made available online at ICARDA web site for public.

Outcomes: Facilitate information exchange and data use between the collaborating institutions and for scientific, technical and environmental matters related to plant genetic resources and climate change in WANA region is available.

Output 7: *Increased capacity of NARS institutions and breeders in WANA region to use molecular markers to breed climate change resilient wheat.*

Target: At least 20 NARS breeders and scientists (disaggregated by gender) from participating countries trained on genetic resources management, breeding and molecular tools using SNP markers and bioinformatics.

Activities:

1. Training courses targeting NARS scientists and breeders to acquire skills in specific areas of molecular breeding.
2. A workshop on the use of web-based tools for breeding activities.

Outcomes: Increased capacity of NARS institutions and breeders on using molecular markers for breeding climate change resilient wheat varieties.

2.4. Targeted PGRFA

The targeted PGRFA in this proposed study is wheat (*Triticum* spp.). Special emphasis will be mainly given on bread (*Triticum aestivum*) and durum wheat (*Triticum durum*) that have potential to harbor new alleles for drought and heat tolerance as inferred by the FIGS approach and molecular work. The used material in the special collection set (~1250 accessions) and the two subsets (bread: ~250 and durum: ~250 accessions) are expected to contain gene bank accessions, prebreeding material, historical and elite breeding lines maintained at ICARDA genetic resources unit and breeding programs. Besides, newly released varieties by ICARDA and NARS of the targeted countries will be used in the implementation of the proposed study. The utilization of wheat germplasm in future breeding activities might result in the production of wheat varieties with improved tolerance against heat and drought that can be used by NARS for wheat improvement and to achieve better food security for poor-resources population groups in WANA region. Any material obtained from this project will be conserved in ICARDA gene bank. Furthermore, phenotypic and genetics information related to the tested material will be made available through the online database at ICARDA web site.

2.5. Direct and indirect beneficiaries

Wheat germplasm identified in this study will directly benefit ICARDA and national partners' wheat breeders in WANA region. Training targeting young scientists and breeders from the selected countries and the development of data sharing tools will allow direct technology transfer and wider access to information that will help breeders improve wheat production in WANA region. The training of breeders, scientists and extension personnel will ensure equality and gender representation. We are expecting that at least 20 NARS breeders and scientists will directly benefit from the training courses and workshops.

In addition, the scientific community will have access to the results from this project that will facilitate the utilization of the characterized wheat germplasm for other studies. We anticipate that hundreds of breeders and wheat scientists will access the web site on annual basis

The utilization of climate resilient wheat germplasm to develop improved drought and heat tolerant cultivars is expected to contribute to enhanced wheat production, food security in targeted countries and to the improvement of the livelihood of tens of thousands poor-resources farmers in WANA region. The results will be presented to the NARS directors in the targeted countries to promote the adoption of any newly released varieties with heat and drought tolerance in their future plans.

Finally, the poor-resources farmers growing wheat under dryland environments will be the ultimate beneficiaries of the results of this project.

2.6. Impact and impact pathways

The project will contribute to to the achievement of **Millennium Development Goals 1 and 7:**

- *To eradicate extreme poverty and hunger*
- *Ensure environmental sustainability*

2.6.1. Food security and poverty alleviation

As mentioned earlier, one of the main objectives in the proposed study is to develop new wheat lines with improved tolerance to drought and heat stress conditions prevalent in WANA region. The new wheat varieties could be utilized by the NARS and resource-poor farmers in the WANA countries to increase the production of wheat to achieve better food security, reduce large scale importation of grains, and alleviate poverty resulting in better livelihood of poor-communities.

In addition, the utilization of molecular markers in breeding programs will assist in accelerating breeding cycles and precise selection of wheat material for different traits. This will assist in delivering wheat lines tailored with selected alleles and genes for drought and heat stress tolerance, and other important traits such as disease resistance, quality and agronomical performance.

Furthermore, the outputs of the proposed study will align with the strategic food security goals of the local governments for wheat by providing breeders and other plant scientists with the tools they need to deliver to farmers new varieties that are adapted to the climate-related challenges with particular emphasis on drought and heat tolerance. This is further aligned with the CRP3.1 Strategic Initiative 4, Outcome 31.04.01 "Effectiveness of partners to utilize international wheat